

FINAL TERM

Physics 1 [Fall 2023-2024]

**Department of Physics
Faculty of Science & Technology (FST)
American International University-Bangladesh**

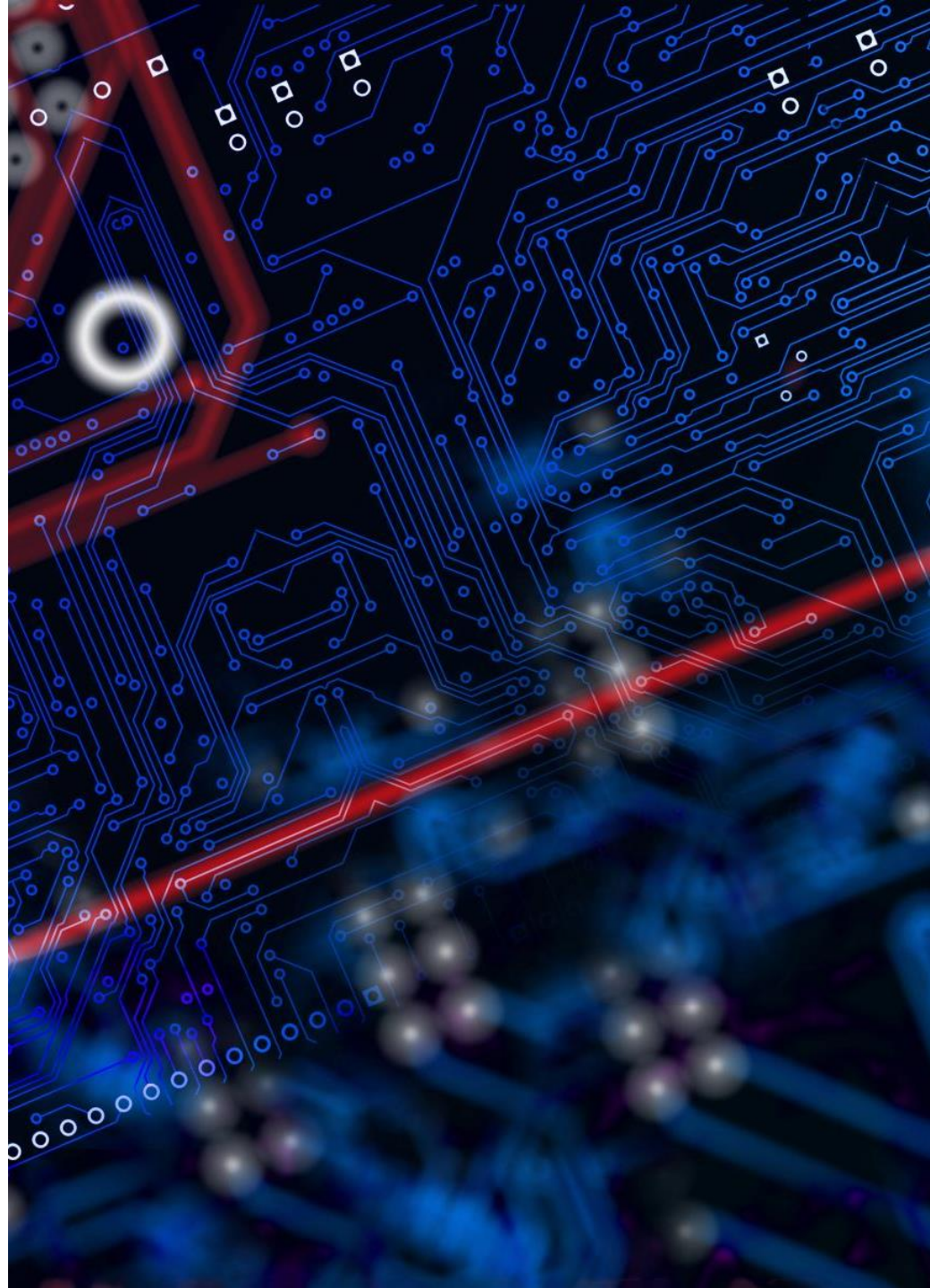
Reference Books

- *Fundamentals of Physics (Edition: 10th)*
Written by Halliday, Resnick and Walker
- *University Physics (13/14th Edition) written*
by Hugh D. Young and Roger A. Freedman

LESSON 1

BOOK CHAPTER 22

ELECTRIC FIELDS



Think, why.....



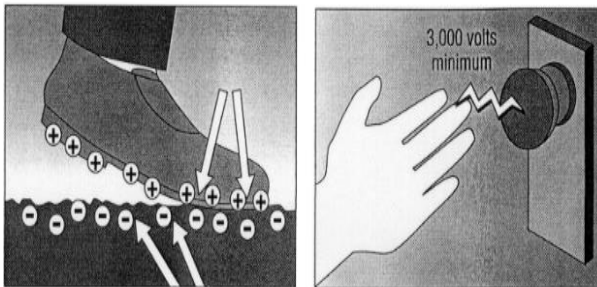
Taking off the hat in the winter



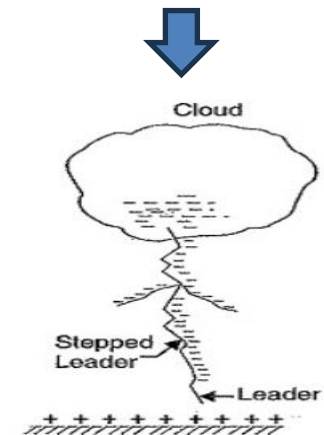
Attraction between the balloon and hair



Lightning



Static charge is responsible for all of these.



Why do you feel the shock when you scuff your shoes across a carpet and then reach for a metal doorknob?



This is due to charged particles leaping between your finger and the doorknob.

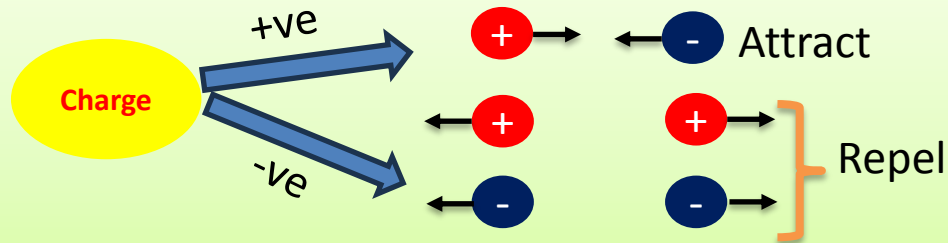


Do you know?

- ❑ The root of the computers, television, radio, telecommunications, household lighting, and even the ability of food wrap to cling to a container etc. depend on the physics of electromagnetism, which is the combination of electric and magnetic phenomena.
- ❑ Electromagnetic interactions involve particles that have a property called *electric charge*.
- ❑ Just as objects with mass are accelerated by gravitational forces, so electrically charged objects are accelerated by electric forces.

Nature of electric charge

Like charges repel, unlike charges attract



- A positive charge and a negative charge attract each other.
- Two positive charges or two negative charges repel each other.

Charge is a conserved quantity.

Net charge $Q = 0$



gains 2 electrons loses 2 electrons

The principle of conservation of charge:
The algebraic sum of all the electric charges in any closed system is constant.

Electric Charge is quantized.

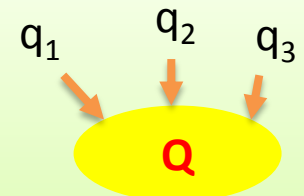
Q

$$Q = n \times e$$

The magnitude of charge of the electron or proton is a natural unit of charge.

Every observable amount of electric charge is always an integer multiple of this basic unit. That is charge is *quantized*

Charges are additive in nature.



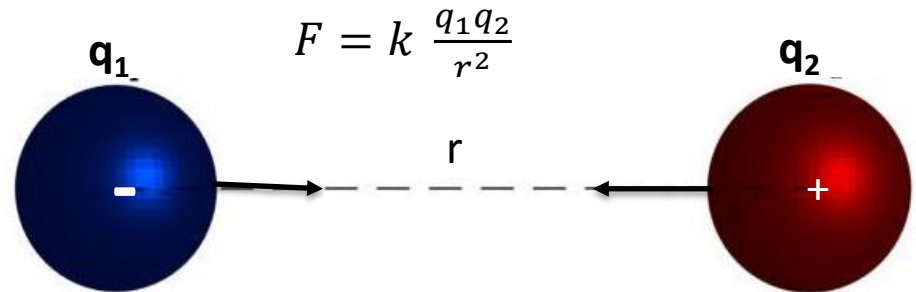
$$Q = q_1 + q_2 + q_3$$

The charge of an electron or a proton,
 $e = 1.6021764871402 \times 10^{-19} \text{ C}$

Coulomb's Force (Electrostatic force)

Coulomb's law: (Charles Augustin de Coulomb)

The magnitude of the electric force between two-point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.



The magnitude **F** of the force that each of two-point charges q_1 and q_2 , a distance r apart, exerts on other can be expressed as

$$F = k \frac{q_1 q_2}{r^2}$$

$$\text{or, } F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\frac{1}{4\pi\epsilon_0} = k = 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$$

$$e = 1.6021764871402 \times 10^{-19} \text{ C}$$

The directions of the forces the two charges exert on each other are always along the line joining them.

Electric force versus gravitational force

An α particle (the nucleus of a helium atom) has mass $m = 6.64 \times 10^{-27}$ kg and charge $q = 3.2 \times 10^{-19}$ C. Compare the magnitude of the electric repulsion between two α ("alpha") particles with that of the gravitational attraction between them.

Answer: The Electric repulsion force between two α particles,

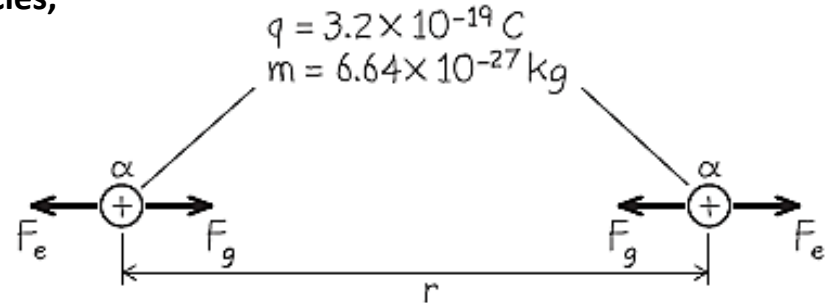
$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$$

The gravitational attractive force between two α particles,

$$F_G = G \frac{m_1 m_2}{r^2} = G \frac{m^2}{r^2}$$

The ratio of electric force to gravitational force,

$$\begin{aligned} \frac{F_e}{F_G} &= \frac{1}{4\pi\epsilon_0 G} \frac{q^2}{m^2} = \frac{9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2}{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{Kg}^2} \frac{(3.2 \times 10^{-19} \text{ C})^2}{(6.64 \times 10^{-27} \text{ Kg})^2} \\ &= 3.1 \times 10^{35} \end{aligned}$$

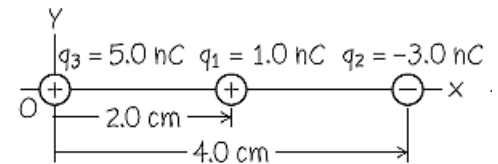
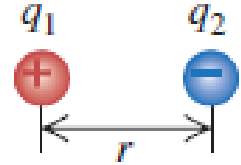


This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force. This is always true for interactions of atomic and subnuclear particles.

Force between two-point charges

Solve it by yourself !

1. Two point charges, $q_1 = 25 \text{ nC}$ and $q_2 = -75 \text{ nC}$ are separated by a distance $r = 3 \text{ cm}$ (Fig. a). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .
2. Two point charges are located on the x-axis of a coordinate system: $q_1 = 1 \text{ nC}$ is at $x = +2.0 \text{ cm}$ and $q_2 = -3.0 \text{ nC}$ is at $x = +4 \text{ cm}$. What is the total electric force exerted by q_1 and q_2 and on a charge $q_3 = 5.0 \text{ nC}$ at $x = 0$?



ELECTRIC FIELD AND ELECTRIC FORCES

Think
about
it....

When two electrically charged particles in empty space interact, how does each one know the other is there?

We can begin to answer this question, and at the same time reformulate Coulomb's law in a very useful way, by using the concept of *electric field*.

ELECTRIC FIELD

(a) *A* and *B* exert electric forces on each other.



To introduce this concept, let's look at the mutual repulsion of two positively charged bodies **A** and **B** (Fig. a). Suppose B has charge q_0 and let F_0 be the electric force of A on B. One way to think about this force is as an **“action-at-a-distance”** force—that is, as a force that acts across empty space without needing any matter (such as a push rod or a rope).

Can you recall that Gravity can also be thought of as an “action-at-a-distance” force or noncontact force?

The repulsion between **A and B** may visualize by considering that (as a result of the charge that A and B carries),

- (1) **Body A**, somehow *modifies the properties of the space around it*.
- (2) Then **body B**, senses how space has been modified at its position. The response of body B is to experience the force, F_0 .

The charged **body A** produces or causes an electric field at point P (and at all other points in the neighborhood). This electric field is present at even if there is no charge at P.

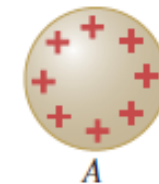
If a point charge q_0 is then placed at point P, it experiences the force, F_0 . We take the point of view that this force is exerted on q_0 *by the field* at P (Fig. c).

Thus the electric field is the intermediary through which **A communicates its presence to q_0** .

(a) *A and B exert electric forces on each other.*

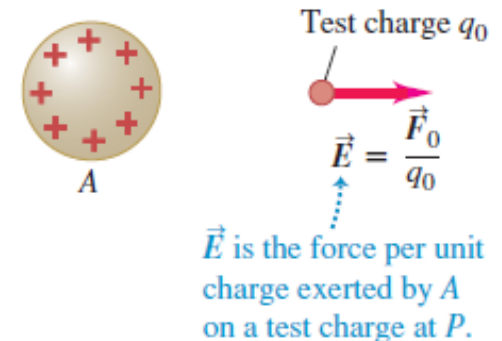


(b) *Remove body B ...*



... and label its former position as P.

(c) *Body A sets up an electric field $\vec{E}$ at point P.*

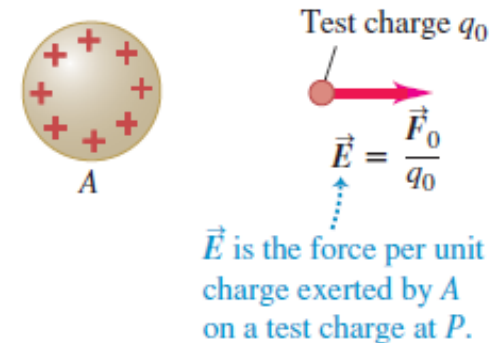


- ❑ A charged body creates an electric field in the space around it.
- ❑ The electric force on a charged body is exerted by the electric field created by *other* charged bodies.

We define the electric field $\vec{E}$ at a point as the electric force $\vec{F}$ experienced by a test charge q_0 at the point, divided by the charge q_0 . That is, *the electric field at a certain point is equal to the electric force per unit charge experienced by a charge at that point:*

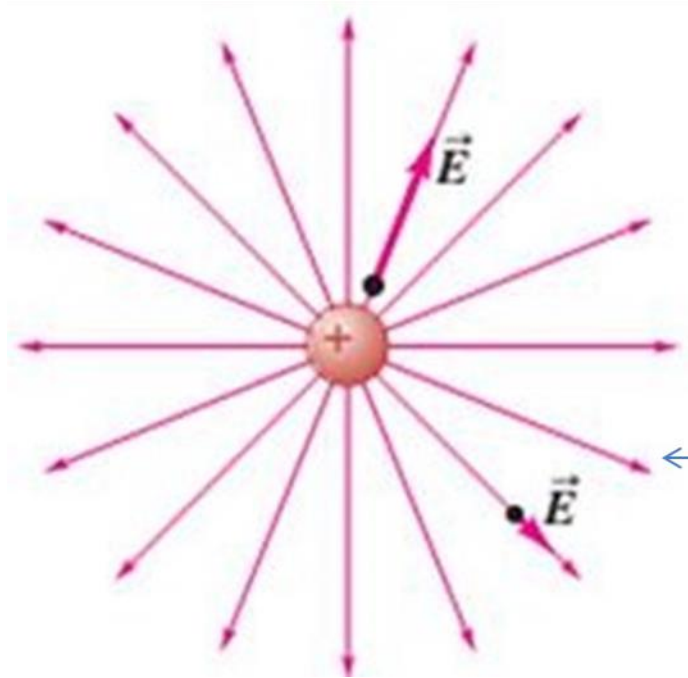
An electric field is a vector field because it is responsible for conveying the information for a force, which involves both magnitude and direction. This field consists of a distribution of electric field vectors $\vec{E}$, one for each point in the space around a charged object.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

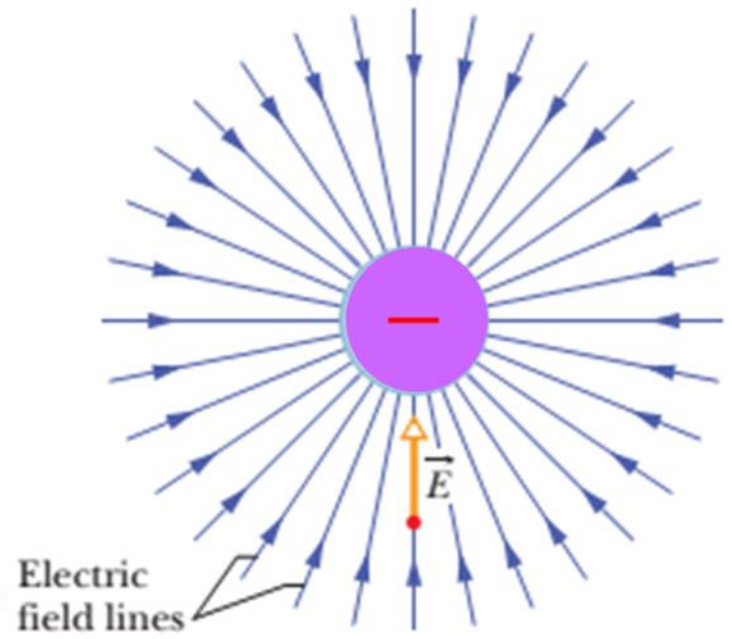


The SI unit for the electric field is the Newton per Coulomb (N/C).

Electric Field lines around charged particles (Point Charges)

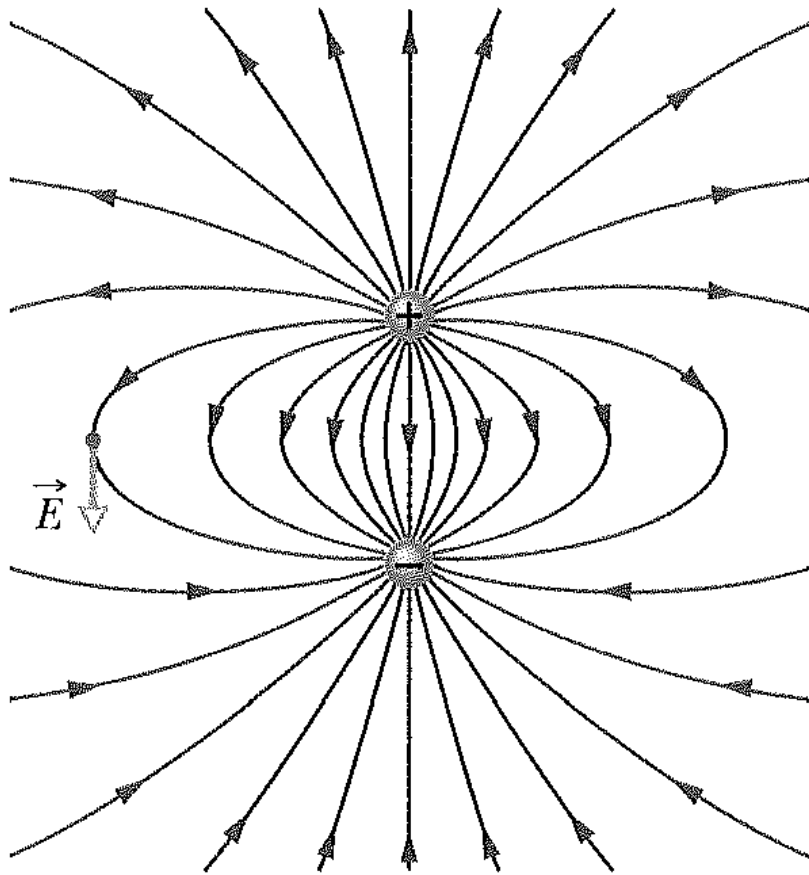


The field produced by **a positive point charge** points away from the charge

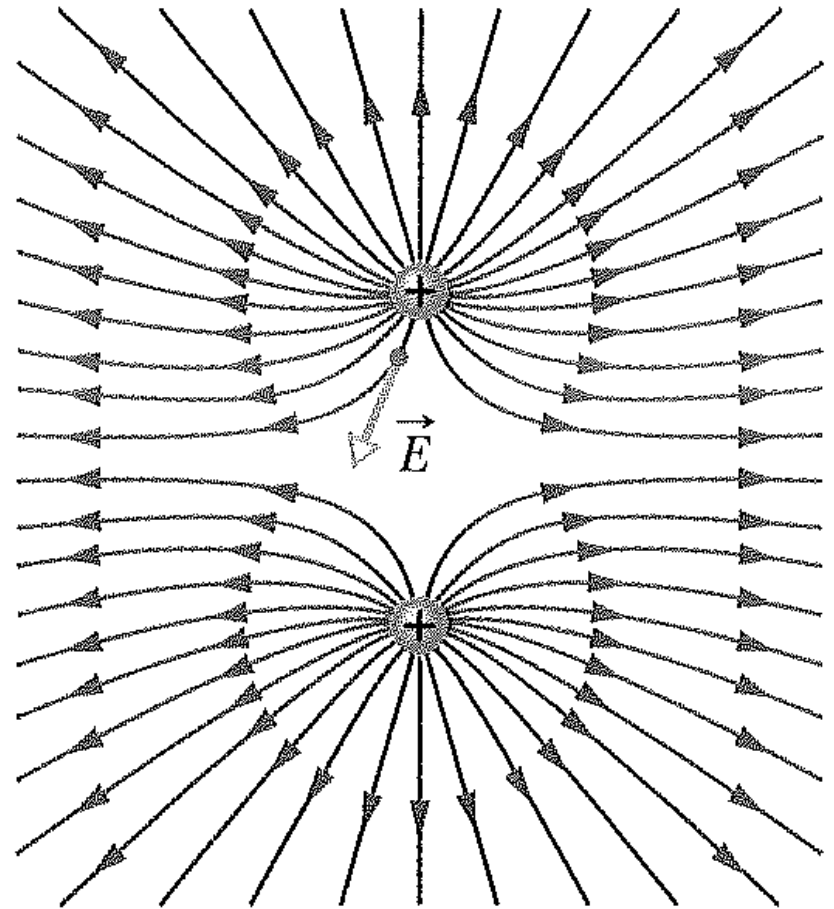


The field produced by **a negative point charge** points toward the charge

Electric field lines extend away from positive charge (where they originate) and toward negative charge (where they terminate).



Field lines for a positive point charge and a nearby negative point charge that are equal in magnitude.



Field lines for two equal positive point charges.

Electric field lines help us visualize the direction and magnitude of electric fields. The electric field vector at any point is tangent to the field line through that point. The density of field lines in that region is proportional to the magnitude of the electric field there. Thus, closer field lines represent a stronger field.

Electric Field due to a Point Charge:

If we place a small test charge $+q_0$ at the field point P at a distance r from the point charge q , the magnitude F of the force is given by the Coulomb's law,

$$F = \frac{1}{4\pi\epsilon_0} \frac{|qq_0|}{r^2}$$

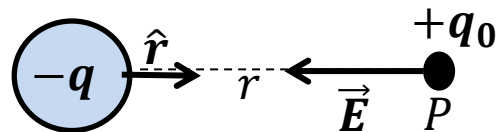
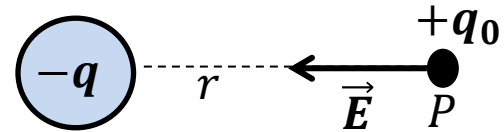
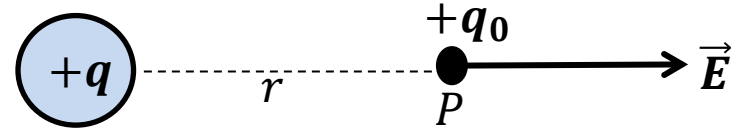
The quantity ϵ_0 , called the **permittivity constant**. The value of ϵ_0 is

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N.m}^2}$$

The magnitude E of the electric field at point P is

$$E = \frac{F}{q_0} = \frac{\frac{1}{4\pi\epsilon_0} \frac{|qq_0|}{r^2}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$



In vector form,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Problem 5 (Book chapter 22)

A charged particle produces an electric field with a magnitude of 2.0 N/C at a point that is 50 cm away from the particle. What is the magnitude of the particle's charge?

Answer: Required formula:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$2 = 9 \times 10^9 \frac{q}{(0.50)^2}$$

$$q = \frac{(2)(0.25)}{9 \times 10^9} = 0.0555 \times 10^{-9} \text{ C}$$

Given

$$E = 2.0 \text{ N/C}$$

$$r = 50 \text{ cm} = 0.50 \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$$

$$q = ?$$

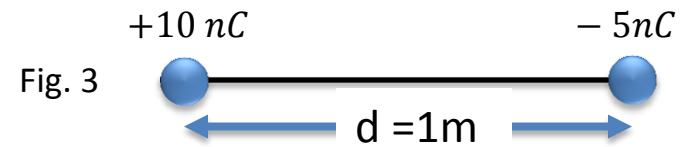
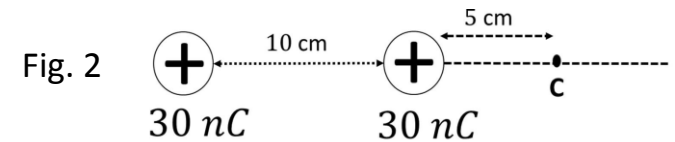
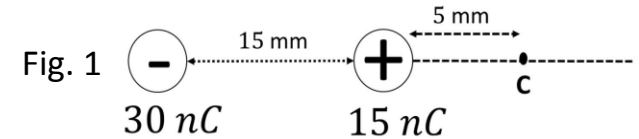
Problem 6 (Book chapter 22)

What is the magnitude of a point charge that would create an electric field of 1.00 N/C at points 1.00 m away?

Answer: Follow exactly similar steps as Problem 5 to answer problem 6.

Let's try !!!!

1. Calculate electric field at point C for figure 1.
2. Calculate net electric field at point C for fig. 2.
3. Calculate net electric field at the mid point of the distance between the two point charges for fig. 3.



THANK YOU